



CIMA STRATEGIC CASE STUDY

Mock Exam 2

Rotomyne

Suggested Solutions

How to use Astranti Solutions

What an Astranti Solution is for

Our solutions are designed to be **perfect answers**. That means that they cover every possible point in depth in order to serve as a learning tool.

This is so that when you review your own solutions against ours, you can see what you got right and what you got wrong. What you covered in enough depth and what you missed out.

What an Astranti Solution is NOT for

Our solutions are not an example of what you should be able to achieve in the exam!

When you sit the real exam, you simply **won't have the time to write an answer as long as ours**. Perhaps you'll cover 75% of it and this is completely fine as long as you have answered the question.

CIMA solutions

You may have noticed our solutions differ from the CIMA solutions. CIMA solutions to past exams are simply examples of a script that might have passed.

They do not prepare, guide or teach you how to improve your answers and increase your chances of passing. Ours do.

Industry examples

You'll notice that we've included real world industry examples throughout the solutions. This is something you should be doing in your own solutions.

Examiners repeatedly emphasise the **importance of wider industry knowledge** and industry examples are the best way to earn the available marks in this area.

Task 1 - Solution

From: Senior Manager Finance

To: Martin Jacobs, Chief Finance Officer (CFO)

Subject: Cloud technology, cyber threats and penetration testing

Introduction

Firstly, I will evaluate the benefits and drawbacks of cloud technology for our organisation. Secondly, I will discuss the possible consequences of the two security attacks and explain how penetration testing could help us prevent such attempts in future.

Sub-task (a) - 40%: Cloud technology

Introduction

Cloud computing enables the storage and access of data over the Internet, facilitating remote access and collaboration. For Rotomyne, which operates in multiple countries and relies on continuous data exchange with customers and suppliers, moving from physical data centres to cloud services offers significant operational advantages. This shift is particularly important as the company looks to diversify into the manufacture of sodium-ion batteries, which will require scalable and secure data infrastructure.

Benefits of cloud technology

On-demand self-service

Cloud technology allows Rotomyne's employees spread across multiple countries, including its mines in politically sensitive regions, to access critical information at any time. This is crucial as Rotomyne manages the real-time exchange of data between its mining operations, factories, and global customers. For instance, production teams can access updated shipping schedules and supply forecasts remotely, ensuring that the fluctuating demands of the electric vehicle industry, one of Rotomyne's largest customer bases, are met without delays. This capability is particularly important as the company begins investing in sodium-ion battery production, where fast, remote decision-making will be critical.

Cost effective

Rotomyne is currently preparing for significant investments, including the development of sodium-ion batteries. By transitioning to a cloud-based infrastructure, the company can reduce its dependency on costly physical data centres in Porrrland and Snowcity, freeing up funds for strategic projects. This shift would also lower IT maintenance costs, vital, as Rotomyne currently lacks a dedicated IT director, allowing the company to focus its financial resources on its core competencies, such as lithium extraction and battery innovation.

Enhanced collaboration

Rotomyne operates across various countries, with facilities ranging from high-tech factories to mining sites in politically unstable regions. Cloud technology would enable instant, real-time collaboration across these geographically dispersed operations. This is especially important as Rotomyne's research and development (R&D) into sodium-ion batteries will require close coordination between its research facility, production plants, and external academic collaborators. For example, the R&D team could collaborate seamlessly with the factory managers to implement new production processes for the batteries without any logistical delays.

Specialist knowledge not required

Given Rotomyne's rapid expansion into new technologies, such as sodium-ion batteries, and its global reach, managing its own IT infrastructure would be burdensome. Outsourcing cloud technology means that technical maintenance and security can be handled externally, a crucial advantage since the company does not have a dedicated IT director. This approach would allow Rotomyne's leadership team to focus on the critical tasks of managing its diversified global operations and addressing the environmental and political challenges its mining activities face.

Back-up and recovery

Rotomyne's largest customer, a vehicle manufacturer, accounts for 22% of its revenue. The company cannot afford disruptions in its operations that could delay shipments of battery-grade lithium hydroxide or lithium carbonate. Recent cyber-attacks targeting the company's systems highlight the vulnerabilities of its current data centres. By moving to cloud storage, Rotomyne can reduce the risk of physical damage to its systems and ensure faster recovery if another cyber-attack occurs, thereby protecting sensitive customer information and avoiding costly delays in production.

Security

While cloud technology generally offers stronger security than traditional physical data centres, it is particularly beneficial for Rotomyne, which operates in regions where physical security is a concern. The company's mines in politically unstable areas are vulnerable to theft, sabotage, and data breaches. Enhanced cloud security could safeguard not only operational data but also sensitive geological and proprietary mining data, ensuring that even if there is physical disruption at a mining site, the company's intellectual property and production processes remain secure.

Scalability and flexibility

With the diversification into sodium-ion batteries, Rotomyne will need to scale its operations rapidly to meet the anticipated rise in demand for these batteries by 2030. Cloud technology provides the flexibility needed to add storage capacity or expand IT capabilities quickly. This ability to scale on demand is crucial as the company looks to expand its customer base beyond traditional vehicle manufacturers into new industries that require energy storage solutions. The ability to scale IT infrastructure without large capital investments will give Rotomyne the agility it needs to remain competitive.

Drawbacks of cloud technology

Security breaches

For Rotomyne, a breach in cloud security could be catastrophic. If its cloud provider were compromised, the company risks having its R&D data on sodium-ion batteries stolen. This is a critical concern, as the success of this new venture hinges on maintaining the confidentiality of its research findings, especially as competitors race to develop similar technologies. Additionally, cloud providers that host multiple clients increase the risk of shared infrastructure attacks.

Regulatory risk

Rotomyne operates in over 90 countries, with each potentially having different data protection laws. Ensuring that a cloud provider complies with these diverse regulations, such as the GDPR in Europe or stricter local rules in certain countries, is a complex task. Failure to comply with these regulations could expose Rotomyne to hefty fines and legal challenges. With mines located in politically sensitive regions, any data breach involving operational or financial data could further complicate relations with local governments and lead to regulatory scrutiny.

Confidentiality issues

Rotomyne holds long-term contracts with key customers, including its largest vehicle manufacturer client. Confidentiality of these contracts is essential, and a cloud provider could expose this information, either through internal access by the provider's employees or via security vulnerabilities. Given the competitive nature of the lithium market, losing control over customer agreements or supply chain information could damage Rotomyne's negotiating power and market position.

Conclusion

While cloud technology offers numerous benefits to Rotomyne, particularly in terms of scalability and collaboration, the potential drawbacks, especially around security breaches and regulatory compliance, cannot be ignored. Rotomyne must weigh these risks carefully, particularly as it diversifies into the sodium-ion battery market, and take steps to implement stringent security protocols and robust contractual safeguards with its cloud provider.

Sub-task (b) - 60%: Consequences of cyber attacks and penetration testing

1bi - Consequences of the two cyber security attacks

The recent system and physical security breaches expose serious vulnerabilities within Rotomyne's infrastructure. Though immediate operational disruptions were avoided, the broader potential consequences of these attacks could have long-lasting impacts on business continuity, data integrity, and the company's reputation.

Threat to information - Theft, disclosure or alteration

One of the most critical risks is the potential theft of sensitive information. Rotomyne holds valuable proprietary data, including research into sodium-ion batteries and strategic contracts with key customers. If this information were accessed, it could significantly harm Rotomyne's competitive advantage, particularly as competitors in the lithium and battery industries are advancing rapidly. Intellectual property theft could weaken Rotomyne's leadership position in these sectors.

Moreover, the threat extends to potential disclosure of confidential customer data. Although it appears that no customer-related Personally Identifiable Information (PIFI) was compromised, the nature of the attack suggests that future breaches could target

such data. If customer data were leaked, it would lead to a loss of trust, damaged relationships, and possible legal actions, which would be especially damaging given Rotomyne's global reach and high-profile clientele.

System disruption and operational delays

Beyond data theft, operational disruptions are another significant concern. Rotomyne's production and shipping operations are heavily reliant on the seamless integration of its Warehouse Management System (WMS) and Industrial Control Systems (ICS). Any prolonged attack that disrupts these systems could delay shipments, especially of lithium hydroxide and lithium carbonate, both of which are critical for fulfilling customer orders. In a competitive market, such delays could result in cancelled orders or reduced confidence from customers who rely on Rotomyne's timely delivery.

Physical security risks

The physical aspect of the attack is equally concerning. Intruders at the Yeeland factory were able to bypass certain security measures, suggesting weaknesses in Rotomyne's physical security protocols. If attackers were to access or disrupt the cables carrying data between critical systems, it could lead to widespread operational chaos. Strengthening physical security measures at key locations, especially those integral to production, is crucial to prevent further breaches.

Legal and regulatory implications

Should customer data have been accessed during these attacks, Rotomyne could face significant legal consequences. With operations spanning over 90 countries, the company must comply with various data protection regulations, including GDPR in Europe. Failing to protect sensitive information could lead to substantial fines and reputational damage. While PIFI was not accessed in this instance, future breaches may not be as fortunate, and regulatory bodies may scrutinise Rotomyne's cybersecurity practices.

Damage to reputation and customer confidence

Rotomyne's ability to maintain trust with its customers and stakeholders depends heavily on its reputation for reliability and security. Even the suggestion of a security breach, especially when it involves physical and digital attacks, can shake investor and customer confidence. If customers perceive Rotomyne as unable to protect sensitive data or secure its operations, they may seek alternative suppliers. This would be particularly damaging in the competitive lithium market, where long-term contracts are essential for sustained revenue.

Impact on share price

The financial markets are highly sensitive to cybersecurity incidents. While no ransom demands have been made thus far, the news of two separate security breaches within a short time frame could lead to uncertainty among investors. If Rotomyne fails to demonstrate that it has taken decisive action to prevent future incidents, the company may experience a drop in share price. A loss of investor confidence could also complicate efforts to secure financing for new projects, such as the proposed expansion into sodium-ion battery production.

Need for immediate response

To mitigate the fallout from these incidents, Rotomyne must act swiftly. Communicating openly with customers, investors, and regulatory bodies will be key to maintaining trust. Additionally, a thorough investigation of both the physical and digital breaches should be conducted, with results shared transparently to demonstrate the company's commitment to addressing these vulnerabilities.

1bii - Penetration testing

Penetration testing (pen testing) allows Rotomyne to simulate attacks on its IT infrastructure, identifying weaknesses before they can be exploited by external actors. This method of proactive testing is essential given the recent breaches, as it provides insights into vulnerabilities that could disrupt both digital and physical operations. Penetration testing consists of three core stages: network discovery, vulnerability probing, and exploiting vulnerabilities.

Network discovery

The first step in penetration testing is network discovery, which maps out the full extent of an organisation's network, including connections between various systems. For Rotomyne, this step is crucial given the breadth of its operations, spanning from its head office in Porland to mines and manufacturing facilities across six countries. The recent cyber attack exploited the interface between the Industrial Control System (ICS) and IT network, revealing vulnerabilities in the connections between these systems. Network discovery could help map out these connections more effectively and assess whether similar vulnerabilities exist elsewhere in the network.

The interaction between IT and OT (Operational Technology) systems, such as the ICS, is particularly concerning. Given that Rotomyne's ICS was developed 20 years ago without considerations for modern cybersecurity threats, the integration of IT systems has

created new entry points for potential breaches. Understanding these weaknesses will allow Rotomyne to focus on strengthening security in critical areas like data cables and external IT resources, such as the junction box targeted in the recent physical attack at Yeeland.

Vulnerability probing

Once the network is mapped, the next stage involves vulnerability probing, where specific devices and systems are tested to identify potential weaknesses. This stage would focus heavily on the Operational Technology (OT) systems within Rotomyne's operations, particularly the ICS, which remains highly vulnerable. Because the ICS was originally built to focus solely on efficiency and safety in mining and manufacturing, cybersecurity concerns were not considered during its development.

Given that the ICS now connects to Rotomyne's broader IT network, it represents a significant vector for potential attacks. Pen testing would reveal how easily unauthorised users could exploit this interface, highlighting potential threats such as unauthorised data access or manipulation of production processes. Additionally, testing could assess the security of network devices used in both digital and physical infrastructure, such as the compromised data cables at the Yeeland facility.

Penetration testing should involve continuous assessments using specialist software to uncover previously unknown vulnerabilities in systems and devices. This process will be vital for understanding the full extent of weaknesses within Rotomyne's network and determining which systems or areas of the company require immediate attention.

Exploiting vulnerabilities

The final stage in penetration testing involves simulating what would happen if identified vulnerabilities were exploited by an external attacker. For Rotomyne, hiring an ethical hacker ('White Hat' hacker) to conduct these simulations would offer invaluable insights into the real-world consequences of such attacks. By attempting to exploit flaws in the ICS and IT network integration, the hacker could determine how easily sensitive information—such as customer contracts or proprietary research—could be accessed or disrupted.

During this phase, the ethical hacker could also simulate the installation of malware or ransomware to test the company's resilience to such attacks. This controlled test would help Rotomyne understand the potential financial and non-financial impacts of these vulnerabilities, including downtime, loss of revenue, or the potential for reputational damage if customer data were compromised.

Simulating these attacks would provide actionable insights into the severity of the risks posed by the current IT and OT infrastructure. The results would allow Rotomyne to develop targeted strategies for mitigating these risks, whether through strengthening security protocols, patching vulnerabilities, or investing in system upgrades.

Actionable insights for Rotomyne

Penetration testing would give Rotomyne a comprehensive understanding of the specific vulnerabilities present in its digital and physical security systems. By addressing these vulnerabilities proactively, the company could reduce its exposure to future attacks. With facilities spread across multiple countries, each with varying levels of security, regular pen testing would ensure that all locations, not just critical hubs like the Yeeland factory, are secured. Furthermore, identifying vulnerabilities in how the ICS interacts with other parts of the company's network could lead to necessary upgrades that protect both proprietary data and operational efficiency.

In addition, pen testing will help Rotomyne prioritise its security investments, focusing on areas of highest risk. For example, securing the interface between IT and OT systems, and improving the protection of external IT infrastructure like data cables, would immediately strengthen the company's defences. Understanding where the company is most vulnerable will also allow it to tailor employee training and response protocols, further reducing the risk of successful attacks in the future.

Task 2 - Solution

From: Senior Manager Finance

To: Martin Jacobs, Chief Finance Officer (CFO)

Subject: Ransomware threat, change in dividend policy and scrip dividend

Introduction

Firstly, I will explain and evaluate the likely outcome if the ransom is paid, but ransomware is not triggered. Secondly, I will discuss the implications of change in dividend policy and evaluate the benefits and drawbacks of a scrip dividend for Rotomyne.

Sub-task (a) - 40%: Ransomware threat

Data theft and damage

One of the primary concerns for Rotomyne, given its recent cyber-attacks, is the potential theft or manipulation of sensitive data. The company's proprietary research on sodium-ion batteries, contracts with global customers, and industrial processes are high-value targets for malicious actors. Should hackers gain access to these assets, it could weaken Rotomyne's competitive position in the market, especially at a time when competitors are rapidly advancing in lithium and battery technology. Additionally, even if the ransomware is not triggered, the attackers could copy sensitive information and threaten to leak it, causing reputational damage and potential legal consequences.

Ransomware

Ransomware attacks aim to encrypt a company's data and systems until a ransom is paid. However, in Rotomyne's case, if the ransom is paid without the ransomware being triggered, it could signal to other hackers that the company is willing to pay, encouraging further attempts. This risk is particularly high given Rotomyne's reliance on uninterrupted production and its vast IT infrastructure spread across different regions. As Rotomyne is involved in critical industries like lithium production, it may be perceived as a lucrative target, with hackers potentially making escalating demands.

The 2017 NHS ransomware attack in the UK is a relevant example, where ransomware caused significant operational disruption, costing the NHS £73 million. Though Rotomyne has not yet faced such operational halts, paying the ransom may only delay further threats. Attackers may continue to exploit the company's systems, knowing that Rotomyne is vulnerable.

Credibility of Board and share prices

Paying P\$100 million as ransom—roughly 5% of Rotomyne's projected profits—would have significant financial and reputational consequences. While the Board may justify the payment to ensure business continuity, it could face scrutiny from shareholders once this payment is reflected in the company's accounts. Profits for 2024 have already fallen by 22%, and with increased R&D spending on sodium-ion batteries, further reductions in profit margins are likely. This ransom payment may exacerbate concerns about Rotomyne's ability to manage its finances effectively.

Shareholders could question the Board's judgement, especially if the ransomware is never triggered. The perception that the Board was "tricked" into paying the ransom could damage their credibility, and investor confidence may decline. In such a scenario, Rotomyne's share prices could fall, complicating its efforts to raise finance for future projects. Investors may view the company as a risky proposition, particularly as it plans to take on additional debt to fund its diversification strategy.

Future threats and attacks

Paying the ransom could also set a dangerous precedent. If attackers believe that Rotomyne is willing to pay large sums, they may continue targeting the company with increasingly sophisticated threats. The company would be forced to dedicate even more resources to IT security, which could become disproportionately expensive. Rotomyne might also face more frequent and harder-to-detect cyber-attacks, making it difficult to assess the credibility of each threat. This would further strain Rotomyne's operational and financial resources.

Additionally, cybercriminals could exploit other vulnerabilities in Rotomyne's systems, such as its Industrial Control System (ICS) or Warehouse Management System (WMS), which are vital for maintaining production and delivery schedules. If these systems were disrupted, it could cause delays in fulfilling customer orders, potentially leading to contract breaches and financial penalties.

Ethical considerations

From an ethical standpoint, paying the ransom could be seen as morally wrong, particularly if the company did not explore alternative actions, such as consulting cybersecurity experts or negotiating with the attackers. Shareholders, regulators, and the wider public might question why the Board did not involve law enforcement agencies or follow legal channels, such as reporting the attack to authorities. This could lead to reputational damage, not only for Rotomyne but also for its executive and non-executive

directors, who might find it harder to be re-elected or secure future Board positions elsewhere.

Moreover, regulators may investigate whether Rotomyne took all necessary precautions to prevent the breach. Failure to report the ransomware attack could lead to legal consequences, particularly if customer data or Personally Identifiable Financial Information (PIFI) was accessed. Even if the ransomware is not triggered, the company's decision to pay could raise questions about its overall cybersecurity strategy and governance practices.

Sub-task (b) - 60%: Dividend policy

2bi - Implications of change in dividend policy

In light of Rotomyne's plan to diversify into the manufacture of sodium-ion batteries, the CEO has suggested reducing or halting dividends for the current financial year. This proposal has significant implications for shareholders, the company's financial structure, and its relationships with lenders. A key consideration for Rotomyne is whether retaining earnings for reinvestment will generate greater long-term value than continuing to pay dividends.

Impact on shareholders

Rotomyne's shareholders, especially institutional investors like retirement funds, have long relied on regular and high dividend payments. For the year ending September 2024, Rotomyne paid out 89% of its profits as dividends, a notably high proportion compared to its competitor Lithdig, which distributed only 50%. Reducing or eliminating dividends may lead to dissatisfaction among these investors, particularly those who depend on dividends for income. The sudden change in policy could lead to an exodus of income-seeking investors, putting downward pressure on the share price as they sell off their holdings.

Impact on share price

Dividend cuts or suspensions can signal a lack of confidence in a company's future cash flows, leading to a negative reaction from the market. Modigliani and Miller's theory of dividend irrelevance argues that the dividend decision should not affect share price in an ideal market. However, in practice, share prices tend to fall when companies announce dividend reductions. Shareholders may interpret this as a sign that Rotomyne is experiencing financial difficulties or is overly burdened by debt, particularly as the company's gearing ratio is already high at 104%. A fall in share price could also complicate

Rotomyne's efforts to raise additional capital, as investors may perceive the company as riskier.

Balancing debt and dividend expectations

Rotomyne's plan to fund its diversification with debt poses additional challenges for maintaining dividend payments. The anticipated increase in borrowings would nearly double the company's existing debt, raising concerns about financial risk. Investors may expect that such an increase in debt should result in long-term capital growth and higher future dividends. However, the immediate reduction in dividends to fund the R&D required for sodium-ion batteries, which won't be commercially viable until 2030, may frustrate shareholders who prefer short-term returns.

Dividend irrelevancy and capital gains

According to Modigliani and Miller's (M&M) dividend irrelevance theory, shareholders should theoretically be indifferent to the dividend decision as long as the diversification strategy into sodium-ion batteries is expected to generate positive Net Present Value (NPV). If this is the case, the value of Rotomyne's shares should rise over time, compensating shareholders with capital gains rather than dividends. However, in practice, markets often react negatively to dividend cuts. Many investors, especially those seeking regular income, may prefer a steady dividend stream and might not be satisfied with future capital gains alone. This can lead to short-term volatility in share price, even if long-term growth prospects are positive.

Implications for Lenders

Lenders will scrutinise Rotomyne's dividend policy as part of their risk assessment, especially since the company's current gearing ratio is already high. If the company takes on additional debt to finance its diversification, lenders may impose restrictive covenants, limiting Rotomyne's ability to pay dividends or requiring it to maintain certain gearing levels. For example, Rotomyne's gearing ratio (currently at 104%) could double if the new debt is approved, pushing it to nearly 2:1. In comparison, Lithdig, with a gearing ratio of 34%, is seen as a lower-risk competitor. Lenders may require Rotomyne to retain more earnings to reduce its gearing and improve financial stability before approving further loans.

Retained earnings and financial flexibility

Reducing or eliminating dividends would allow Rotomyne to retain more earnings, strengthening its equity base and potentially improving its ability to negotiate with

lenders. Increasing retained earnings could reduce the company's reliance on external financing and improve its balance sheet, creating a more stable financial foundation for future growth. However, given the high levels of debt already on the balance sheet, the impact of retaining earnings may not be sufficient to offset the increased financial risk from doubling the company's debt load.

Risk of shareholder exit

Should Rotomyne reduce dividends, it runs the risk of losing shareholders who are primarily income-seeking. These shareholders may prefer to exit their positions in favour of investments that offer more reliable dividend streams. This could exacerbate volatility in the company's share price and reduce liquidity in the market for Rotomyne's shares. For a company with a beta of 1.63, indicating higher volatility compared to the market, even minor changes in dividend policy could trigger significant fluctuations in share price.

Conclusion

The decision to cut or suspend dividends presents both risks and opportunities for Rotomyne. On one hand, retaining earnings will provide the financial flexibility needed to invest in high-potential projects like sodium-ion batteries. On the other hand, the immediate consequences—such as a potential drop in share price, shareholder dissatisfaction, and the perception of increased financial risk—must be managed carefully. The company's ability to communicate the long-term value of its diversification strategy will be critical in maintaining investor confidence and ensuring that any short-term volatility is mitigated by future growth.

2bii - Scrip dividend and a discussion of its benefits and drawbacks

A scrip dividend allows a company to issue additional shares to its shareholders instead of paying cash dividends. For Rotomyne, which is seeking to preserve cash for reinvestment into its diversification into sodium-ion battery production, this approach could help conserve resources while still offering shareholders value. However, there are both advantages and disadvantages to consider before implementing this strategy.

Advantages - No change in ownership proportion

One of the primary advantages of a scrip dividend is that it does not dilute shareholders' ownership stakes in the company. Each shareholder receives additional shares proportional to their existing holdings, maintaining their ownership percentage in Rotomyne. This ensures that shareholders do not lose any control or influence in the company. While the number of shares increases, the total ownership structure remains

the same. For long-term shareholders, this may be seen as a positive, as they retain their influence while the company reinvests cash for future growth.

Advantages - Cash retention for reinvestment

The main appeal of a scrip dividend for Rotomyne lies in its ability to conserve cash. With significant capital required for the company's diversification into sodium-ion batteries, retaining cash rather than distributing it to shareholders allows Rotomyne to reinvest in R&D and production. This is especially important given that the new product line is not expected to be commercially available until 2030, requiring sustained investment over several years. Issuing a scrip dividend means Rotomyne can build its future capacity without weakening its cash position, making it easier to manage the financial risks associated with the project.

Advantages - Tax benefits for shareholders

From a shareholder perspective, scrip dividends can provide tax advantages. In many jurisdictions, selling shares received through a scrip dividend is subject to capital gains tax rather than income tax, and this could be beneficial depending on individual tax rates. Shareholders may also avoid stamp duty and commission charges that would have been incurred if they had purchased the shares directly on the open market. This can enhance the attractiveness of the scrip option, especially for shareholders seeking to maximise their post-tax returns.

Impact on share price and share capital

The issuance of scrip dividends will increase Rotomyne's share capital, potentially diluting the value of each share in the short term as the same company value is spread across more shares. However, this increase in share capital can improve liquidity and trading volume, making the stock more attractive to investors in the long run. If Rotomyne successfully grows its sodium-ion battery business, the increase in shares could ultimately enhance market capitalisation as investor confidence improves. In the long term, the company could experience greater trading activity and higher valuations if the diversification proves profitable.

Disadvantages - Non-distributable capital

One downside of a scrip dividend is that once retained earnings are converted into share capital, they are no longer distributable as cash dividends. For Rotomyne, this means that reserves used for the scrip dividend would be locked in as share capital and could not be used for other purposes, such as future cash payouts to shareholders. This reduces the

company's flexibility in managing its finances, particularly if it faces unforeseen financial challenges in the future, such as further investments in R&D or mitigating the impacts of market fluctuations in lithium prices.

Disadvantages - Negative market signals

Issuing a scrip dividend could send a negative signal to the market, particularly if shareholders interpret the decision as an indication that Rotomyne is facing cash flow difficulties. While the company is currently managing its resources to fund a large capital project, the perception that it cannot afford to pay cash dividends might undermine investor confidence. This is especially concerning given Rotomyne's current financial position, with a high gearing ratio and increased debt levels as the company funds its diversification strategy. A drop in share price could occur if investors view the scrip dividend as a sign of financial strain.

Disadvantages - Impact on shareholder wealth and ratios

Although scrip dividends may preserve cash, they do not generate any economic value for shareholders in the short term. The overall wealth of shareholders remains unchanged, as the value of each share decreases in proportion to the increase in the number of shares issued. Additionally, issuing scrip dividends has no impact on Rotomyne's financial ratios, such as return on equity or debt-to-equity ratio, which could limit the perceived benefit of the move. This could be seen as a balance sheet adjustment rather than a meaningful financial strategy for improving shareholder returns.

Conclusion

In conclusion, while a scrip dividend offers Rotomyne a way to preserve cash for its ambitious diversification strategy, there are potential downsides, particularly around the market perception of the company's financial health. Careful communication with shareholders will be essential to ensure that they understand the long-term benefits of the strategy and do not interpret it as a signal of cash flow problems. If executed correctly, a scrip dividend could help Rotomyne retain resources for growth while still providing shareholders with value, positioning the company for future success.

Task 3 - Solution

From: Senior Manager Finance

To: Martin Jacobs, Chief Finance Officer (CFO)

Subject: Internal Controls and Corporate Governance

The issue

Firstly, I will recommend and justify four internal controls (other than penetration testing) that we might introduce to prevent the recurrence of this threat. Secondly, I will evaluate arguments for and against the appointment of an additional non-executive director with a background in cybersecurity.

Sub-task (a) - 50%: Internal controls

Introduction

According to IBM, it takes an average of 206 days to detect a cyber attack. In Rotomyne's case, although our SIEM system flagged a minor irregularity, the true disruptive nature of the incident was not identified until the ransomware threat emerged. Many cyber attacks operate covertly, collecting data rather than directly disrupting systems. Therefore, it is critical to implement robust internal controls that continuously monitor network activity and detect suspicious behaviour before any serious damage occurs.

Below are four recommended internal controls that could be introduced to prevent a recurrence of this type of attack.

1. Establishing a Security Operations Centre (SOC) and appointing a CISO

Rotomyne currently uses a SIEM (Security Information/incident and Event Management) system to monitor its IT infrastructure. While SIEM is helpful for detecting anomalies, the SQL injection attack indicates that Rotomyne needs a more comprehensive, proactive defence. One solution is to establish a Security Operations Centre (SOC), a dedicated unit responsible for 24/7 monitoring of security threats across the company's entire IT environment.

A SOC would not only monitor and investigate security incidents in real time but also help coordinate responses to cyber attacks before they cause significant damage. Given the sophistication of recent threats, outsourcing this function to a trusted cybersecurity

firm—such as the external consultant who successfully removed the malware—would provide access to specialised skills and tools that Rotomyne currently lacks.

In addition to the SOC, appointing a Chief Information Security Officer (CISO) would ensure there is dedicated leadership for managing cybersecurity risks. A CISO would work closely with the SOC to assess threats, review security protocols, and develop long-term strategies to mitigate risks. Having this expertise at the senior level would ensure that cybersecurity remains a priority in Rotomyne's strategic decisions.

2. Restricting staff access and enhancing physical security

The recent cyber attack revealed weaknesses in Rotomyne's access control, particularly regarding the phishing email that led to the SQL injection. To mitigate internal risks, it is essential to impose stricter physical and digital access controls across the organisation.

On the physical side, biometric identification systems should be implemented to limit access to critical areas, such as data centres and server rooms. Only authorised personnel should have physical access to these spaces, reducing the risk of unauthorised tampering with hardware or systems. For instance, limiting access to key infrastructure at high-risk sites like the Yeeland factory, which supplies over 60% of Rotomyne's products to its largest customer, would significantly reduce exposure to potential insider threats.

For digital access, role-based access control (RBAC) should be enforced. This would ensure that employees only have access to the systems and data necessary for their specific roles. In the case of the Senior Inventory Manager at Yeeland, their access should have been restricted to inventory management functions, without access to sensitive systems that could be compromised through an SQL injection. Implementing two-factor authentication (2FA) for system access would further strengthen security, ensuring that even if an employee's credentials are compromised, attackers cannot easily gain access to critical systems.

3. Comprehensive staff training and email security

The recent incident likely occurred due to a lack of staff awareness about phishing attacks and social engineering. To prevent similar breaches in the future, we must introduce a comprehensive staff training programme that educates employees on the latest cybersecurity threats, including phishing and malware distribution.

Training should be tailored to different roles within the company. Staff who have direct access to sensitive data, such as inventory managers, should receive specialised training on recognising suspicious emails, identifying potential threats, and understanding the consequences of human error. Regular, mandatory training sessions would help ensure

that all employees are up to date on the latest cybersecurity practices. Additionally, we could collaborate with key suppliers and customers to implement validation checks for emails and communications, reducing the likelihood of employees falling victim to phishing scams.

In addition to training, Rotomyne should enhance its email security by introducing filtering systems that automatically flag suspicious emails. Implementing email verification protocols, especially for emails from external sources, would help prevent staff from interacting with malicious messages. Ensuring that all emails go through robust filtering will reduce the likelihood of successful phishing attempts and protect the company's critical systems.

4. Complete IT system overhaul based on ISO 27001 standards

Rotomyne's Industrial Control System (ICS), which has been in use for over 20 years, presents a significant vulnerability due to its age and the lack of modern security features. To mitigate the risks posed by this outdated system, Rotomyne should undertake a complete IT system overhaul based on the internationally recognised ISO 27001 standards.

ISO 27001 provides a framework for developing and maintaining an Information Security Management System (ISMS). By following this framework, Rotomyne can identify and mitigate cybersecurity risks across its entire IT infrastructure, ensuring that all systems, including the ICS, are protected from modern threats. This would involve conducting a thorough audit of current systems, identifying areas of vulnerability, and implementing comprehensive controls to protect against these risks.

Additionally, introducing Network Configuration Management (NCM) as part of this overhaul would allow Rotomyne to manage and monitor all hardware and software configurations across its network. NCM would ensure that updates and patches are applied regularly, reducing the risk of vulnerabilities being exploited by attackers. By centralising the management of IT systems and implementing ISO 27001, Rotomyne would significantly enhance its cybersecurity posture.

Sub-task (b) - 50%: Appointment of NED

3bi - Arguments for appointment

Demonstrate commitment to cyber security

Appointing a NED with expertise in cybersecurity would signal to stakeholders that Rotomyne is taking cybersecurity risks seriously. Given the increasing frequency of cyber attacks in the industry, such a move would demonstrate proactive risk management. This would reassure shareholders, customers, and investors that the company is committed to safeguarding its operations against evolving threats. In today's environment, where even minor vulnerabilities can be exploited, this appointment would highlight Rotomyne's forward-thinking approach to managing cyber risks.

Providing specialist expertise to the Board

A NED with cybersecurity experience would bring essential expertise to the Board's discussions. Cybersecurity is a complex and rapidly changing field, and having a specialist on the Board would enhance the Board's ability to critically evaluate Rotomyne's cybersecurity strategies. This NED would provide insights into the latest trends and technologies, ensuring that the company's security measures evolve alongside emerging threats.

Additionally, this NED could contribute to the Board's Risk Committee and Audit Committee, where they would ensure that cybersecurity risks are properly scrutinised and managed in line with other business risks.

Aligning with corporate governance best practices

The UK Corporate Governance Code recommends that at least half of the Board, excluding the Chair, should consist of independent NEDs. Rotomyne's current Board structure is somewhat imbalanced, with five executive directors and only three NEDs. Appointing a cybersecurity-focused NED would help restore this balance, providing greater independence and diverse expertise at the strategic level.

Having an independent NED focused on cybersecurity would also allow the Board to challenge management's decisions more effectively, leading to improved governance and risk management practices.

Enhancing risk management

Cybersecurity is a strategic risk that has the potential to disrupt operations, harm reputation, and cause financial damage. Appointing a cybersecurity expert to the Board would improve Rotomyne's overall risk management. With this expertise in place, the Board would be better equipped to assess how cybersecurity risks could impact the business, particularly as Rotomyne expands into the production of sodium-ion batteries. This would allow the company to adopt a more comprehensive approach to risk management, integrating cybersecurity with other strategic risks.

3bii - Arguments against appointment

Limited role of non-executive directors

While NEDs provide valuable oversight and guidance, their role is primarily strategic rather than operational. A NED with a background in cybersecurity might not have the authority or day-to-day involvement needed to effectively manage cybersecurity threats. Instead, Rotomyne may benefit more from appointing a Chief Information Security Officer (CISO), who would be directly responsible for implementing and managing cybersecurity strategies. A CISO would be more hands-on in addressing Rotomyne's cybersecurity needs, which may be a more practical solution than appointing a NED.

Risk of overemphasis on cybersecurity

Focusing too heavily on cybersecurity at the Board level could detract from other critical business areas. While cybersecurity is an important risk, Rotomyne is also facing significant challenges related to its diversification into sodium-ion battery production. Appointing a cybersecurity expert may inadvertently shift too much attention away from other strategic priorities, leading to an imbalance in risk management.

Potentially negative signal to stakeholders

Appointing a cybersecurity-focused NED could send the unintended message that Rotomyne is expecting further attacks. This could raise concerns among stakeholders, particularly investors, about the company's ability to manage its cybersecurity risks. While the intention is to demonstrate proactive management, the appointment could be perceived as a response to an ongoing crisis, potentially leading to a loss of confidence among shareholders and customers.

Imbalance in Board priorities

While cybersecurity is a growing concern, it is not the only risk that requires Board attention. The appointment of a NED focused solely on cybersecurity could skew the Board's focus and result in other areas being neglected. Rotomyne's Board must maintain a balanced approach, ensuring that operational efficiency, financial health, and strategic initiatives, such as the production of sodium-ion batteries, receive adequate attention alongside cybersecurity.

Recommendation

After evaluating the arguments, the appointment of a cybersecurity-focused NED could certainly bring valuable expertise and demonstrate Rotomyne's commitment to addressing cybersecurity risks. However, the role of a NED is inherently limited to providing oversight, which may not be sufficient to address the company's immediate cybersecurity needs. Additionally, the appointment could send a negative signal to stakeholders, suggesting that Rotomyne is vulnerable to further attacks.

A more practical and impactful solution would be to appoint a Chief Information Security Officer (CISO). A CISO would be directly involved in managing cybersecurity risks, implementing controls, and ensuring that Rotomyne's IT infrastructure is secure. This role would provide the hands-on leadership needed to protect the company from future threats.

The Nomination Committee should carefully consider whether the appointment of a NED with cybersecurity expertise is necessary in addition to a CISO. However, in the short term, prioritising the appointment of a CISO would provide the immediate expertise and oversight required to safeguard Rotomyne's operations without overloading the Board with specialised roles.

Marking Matrix

Task	Topic	Core activity	%	Marks	Total
1a	Cloud technology - Benefits and Drawbacks	A	40	13	33
1b	Cyber attack consequences and penetration testing	B	60	20	
2a	Cyberthreat - Ransomware demand	A	40	13	33
2b	Dividend policy change and scrip dividend	C	60	20	
3a	Internal controls (to prevent cyber attacks)	D	50	17	34
3b	Appointment of additional NED	E	50	17	
					100

Marking guidance

Task 1		
(a)	A. Develop business strategy	40%
(b)	B. Evaluate the business ecosystem and business environment	60%

A. Develop business strategy (13 marks)		40%
1ai - Benefits of cloud technology		8 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-3
Level 2	Some understanding demonstrated of the benefits of cloud technology. Technical clarity, accuracy and/or application to the scenario might be lacking.	4-6
Level 3	Very good understanding demonstrated of the benefits of cloud technology with clear, accurate explanations and detailed application to the scenario.	7-8
1aii - Drawbacks of cloud technology		5 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-2
Level 2	Some understanding demonstrated of the drawbacks of cloud technology. Technical clarity, accuracy and/or application to the scenario might be lacking.	3-4
Level 3	Very good understanding demonstrated of the drawbacks of cloud technology with clear, accurate explanations and detailed application to the scenario.	5

B. Evaluate the business ecosystem and business environment (20 marks)		60%
1bi - Consequences of the cyber attacks		10 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-3
Level 2	Some understanding demonstrated of the consequences of the cyber attacks. Technical clarity, accuracy and/or application to the scenario might be lacking.	4-7
Level 3	Very good understanding demonstrated of the consequences of the cyber attacks with clear, accurate explanations and detailed application to the scenario.	8-10
1bii - Penetration testing and cyber attack prevention		10 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-3
Level 2	Some understanding demonstrated of how penetration testing could help us prevent such attempts in future. Technical clarity, accuracy and/or application to the scenario might be lacking.	4-7
Level 3	Very good understanding demonstrated of how penetration testing could help us prevent such attempts in future with clear, accurate explanations and detailed application to the scenario.	8-10

Task 2		
(a)	A. Develop business strategy	40%
(b)	C. Recommend financing strategies	60%

A. Develop business strategy (13 marks)		40%
2ai - Ransom paid but ransomware not triggered - evaluation of outcomes		13 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-4
Level 2	Some understanding demonstrated of the likely outcomes of ransom paid but ransomware not triggered. Technical clarity, accuracy and/or application to the scenario might be lacking.	5-9
Level 3	Very good understanding demonstrated of the likely outcomes of ransom paid but ransomware not triggered with clear, accurate explanations and detailed application to the scenario.	10-13

C. Recommend financing strategies (20 marks)		60%
2bi - Discussion of the implications of change in dividend policy		10 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-3
Level 2	Some understanding demonstrated the implications of change in dividend policy .Technical clarity, accuracy and/or application to the scenario might be lacking.	4-7
Level 3	Very good understanding demonstrated the implications of change in dividend policy with clear, accurate explanations and detailed application to the scenario.	8-10

2bii - Benefits and drawbacks of scrip dividends		10 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-3
Level 2	Some understanding demonstrated the benefits and drawbacks of scrip dividends .Technical clarity, accuracy and/or application to the scenario might be lacking.	4-7
Level 3	Very good understanding demonstrated of the benefits and drawbacks of scrip dividends with clear, accurate explanations and detailed application to the scenario.	8-10

Task 3		
(a)	D. Evaluate and mitigate risk	50%
(b)	E. Recommend and maintain a sound control environment	50%

D. Evaluate and mitigate risk (17 marks)		50%
3ai - Identify four internal controls to prevent cyber attacks		8 marks
	No rewardable material	0
Level 1	One internal control identified.	1-3
Level 2	Two or three controls identified.	4-6
Level 3	Four controls identified.	7-8
3aii - Explain how internal controls will prevent cyber attacks		9 marks
	No rewardable material	0
Level 1	Controls are generic and not applied.	1-3
Level 2	Controls are explained, but are not fully applied.	4-6
Level 3	All controls are fully explained and applied to the particular situation.	7-9

E. Recommend and maintain a sound control environment (17 marks)		50%
3bi - Arguments for appointment		8 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-2
Level 2	Some arguments made for appointment of an additional non-executive director with a background in cybersecurity. Technical clarity, accuracy and/or application to the scenario might be lacking.	3-5
Level 3	Very good arguments made for appointment of an additional non-executive director with a background in cybersecurity. of with clear, accurate explanations and detailed application to the scenario.	6-8
3bii - Arguments against appointment with recommendation		9 marks
	No rewardable material	0
Level 1	Little or no rewardable material here. Very poor understanding and application demonstrated.	1-3
Level 2	Some arguments made against appointment of an additional non-executive director with a background in cybersecurity. Technical clarity, accuracy and/or application to the scenario might be lacking.	4-6
Level 3	Very good arguments made against appointment of an additional non-executive director with a background in cybersecurity with clear, accurate explanations and detailed application to the scenario.	7-9